INCH-POUND

MIL-DTL-32655
28 May 2020

DETAIL SPECIFICATION

CHAINS AND ATTACHMENTS, CORROSION RESISTANT STEEL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers corrosion resistant steel (CRES) chains and attachments commonly used with chains.

1.1.1 Other chain materials. For chains and attachments made from materials other than stainless steel see RR-C-271, "Chains and Attachments, Carbon and Alloy Steel". This includes some brass, bronze and aluminum items. Beaded chain and attachments in CRES are also in RR-C-271.

1.2 Classification. The chains and attachments are of the following types, classes and styles as specified below.

1.2.1 Chain types. The types and classes of welded chain are as follows:

Type I - Chain, welded (PIN code –C01)

Class 1 – Grade 50 chain
Class 2 – Regular-link chain
Class 3 – Long link chain
Class 5 – Twist-link chain

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1.2.2 Attachment types. Attachment types, grades, and classes are as follows:

Type II – Connecting links (PIN code –A02)

Type IVA – Anchor shackles (PIN code –A41)

Class 1 – Round pin and cotter
Class 2 – Screw pin
Class 3 – Bolt, nut, and cotter

Type IVB – Chain shackles (PIN code –A42)

Class 1 – Round pin and cotter
Class 2 – Screw pin
Class 3 – Bolt, nut, and cotter

Type V – Hooks (PIN code –A05)

Style 1E – Eye Slip hook
Style 1C – Round pin clevis Slip hook
Style 2E – Eye Grab hook
Style 2C – Round pin clevis Grab hook
Style 5S – Swivel Eye Hoist hook
Style 5T – Threaded-shank Hoist hook
Style 5U – Unthreaded-shank Hoist hook

Type VI– Rings (PIN code –A06)

Type VII– Swivels (PIN code –A07)

Class 1 – Chain swivel
Class 2 – Eye and eye swivel
Class 3B – Jaw and eye swivel, bolt, nut and cotter
Class 3C – Jaw and eye swivel, round pin clevis
Class 4B – Jaw and jaw swivel, bolt, nut and cotter
Class 4C – Jaw and jaw swivel, round pin clevis

Type XIII – Pear-shaped link (PIN code –A13)

Type XIV – Mechanical coupling link (PIN code –A14)

Type XVI – Master links (PIN code –A16)

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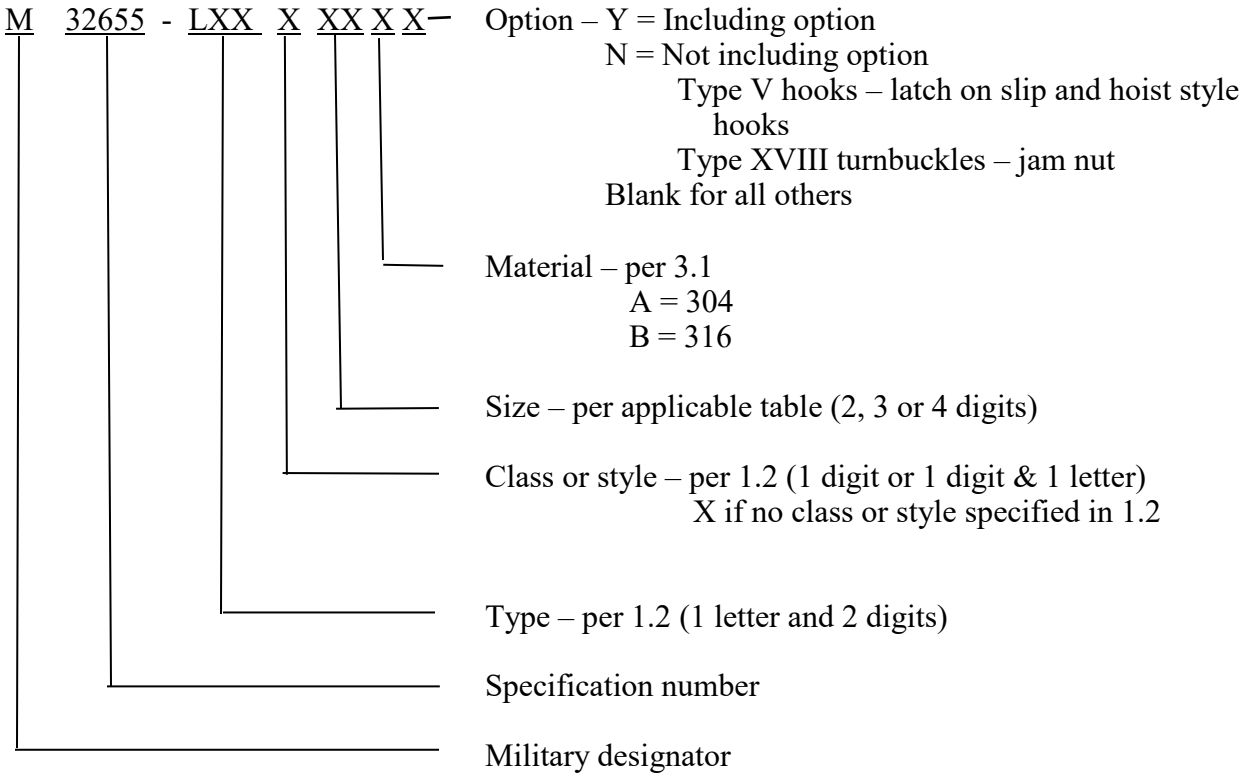
Type XVII – Links, loops and D-rings (PIN code –A17)

Class 1 – Rectangular links
Class 2 – Triangle loop
Class 3 – D – rings
Class 4 – Heavy – duty D – ring

Type XVIII – Turnbuckles (PIN code –A18)

Class 1B – Jaw and jaw turnbuckle, bolt, nut and cotter
Class 1C – Jaw and jaw turnbuckle, round pin clevis
Class 2B – Jaw and eye turnbuckle, bolt, nut and cotter
Class 2C – Jaw and eye turnbuckle, round pin clevis
Class 3 – Eye and eye turnbuckle
Class 4 – Stub and stub turnbuckle
Class 5 – Turnbuckle body

1.3 Part or identification number (PIN). The PIN to be used for chains and attachments acquired to this specification is created as follows:



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Examples: M32655-C01106A – Type I, class 1 (Grade 50) welded chain, 3/8 inch size, 304 or 304L
M32655-A42309B – Type IVB, class 3, bolt/nut/cotter chain shackle, 9/16 inch size, 316 or 316L
M32655-A051E05CN – Type V, class 1E, eye slip hook, 5/16 inch size, 316 or 316L, no latch

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issue of these are those cited in the solicitation or contract.

Military Standard

MIL-STD-1916	- DoD Preferred Methods for Acceptance of Product

(Copies of these documents are available on-line at https://quicksearch.dla.mil)

2.3 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

American Society for Quality (ASQ)

ASQ H1331	- Zero Acceptance Number Sampling Plans

(Copies of this document are available online at https://asq.org/.)

ASME

ASME B1.1	- Unified Inch Screw Threads (UN and UNR Thread Forms)

(Copies of this document are available online at https://www.asme.org/.)

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ASTM International

ASTM A952/A952M	-	Standard Specification for Forged Grade 80 and Grade 100 Steel Lifting Components and Welded Attachment Links
ASTM A967/A967M	-	Standard Specification for Chemical Passivation Treatments of Stainless Steel Parts
ASTM F1145	-	Standard Specification for Turnbuckles, Swaged, Welded, Forged

(Copies of these documents are available at https://www.astm.org.)

SAE INTERNATIONAL

SAE AMS2430	-	Shot Peening, Automatic

(Copies of this documents are available at https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Chains.

3.1.1.1 Type I, welded chains. All classes of type I chain shall be made of CRES 316L (UNS S31603) or 316 (UNS S31600) when 316 is specified, and 304L (UNS S30403) or 304 (UNS S30400) when 304 is specified. The specific alloy is at the manufacturer's option.

3.1.2 Attachments. All attachments and components shall be made of CRES 316 (UNS S31600) or 316L (UNS S31603) when 316 is specified, or 304 (UNS S30400), 304L (UNS S30403) or 347 (UNS S34700) when 304 is specified. The specific alloy is at the manufacturer's option.

3.1.3 Material specification. If a material is not specified in the solicitation or contract (see 6.2), the supplied material is at the manufacturer's option. The PIN shall represent the actual material.

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3.2 Finish.

3.2.1 Chains. Unless otherwise specified (see 6.2), all chain types and classes shall be passivated in accordance with ASTM A967/A967M, glass bead blasted per SAE AMS2430, or by tumbling and/or vibrating with organic material.

3.2.2 Attachments. Unless otherwise specified (see 6.2), all chain types and classes shall be passivated in accordance with ASTM A967/A967M, or glass bead blasted per SAE AMS2430.

3.2.3 Passivation effectiveness. Effectiveness of the passivation treatment shall be confirmed by any test practice listed in ASTM A967/A967M.

3.3 Detail requirements – chains.

3.3.1 General. The dimensions, physical requirements and other detailed data shall be as shown in the respective figures and tables.

3.3.1.1 Working load limit. The working load limit (also shown as working load or WLL) is the maximum load that the chain is rated to carry in non-shock loading conditions.

3.3.1.2 Proof and breaking loads. All chains except twist-link chain shall be proof tested as required at a minimum of 2 times the working load limit. Samples from all lots of all chains shall be break tested as required at a minimum of 4 times the working load limit. Sampling shall be in accordance with 4.3.1. Fixture shall be in accordance with 4.4.1.1.3.

3.3.1.3 Proof test. Chain withstanding the proof load without loss of chain integrity shall have passed the proof test. Marks due to link to link or link to fixture contact shall not be considered as signs of loss of integrity. Links or segments of chain not withstanding the proof load shall be removed.

3.3.1.4 Elongation. All type I chain shall show elongation of not less than 15% after being subjected to the minimum breaking load. Chains are not required to be tested to failure provided the minimum breaking load is exceeded.

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3.3.2 Type I, welded chains.

3.3.2.1 Type I, class 1, grade 50 chain. The welded grade 50 chain shall be similar to figure 1 and shall be in accordance with the dimensions and working loads shown in table I.

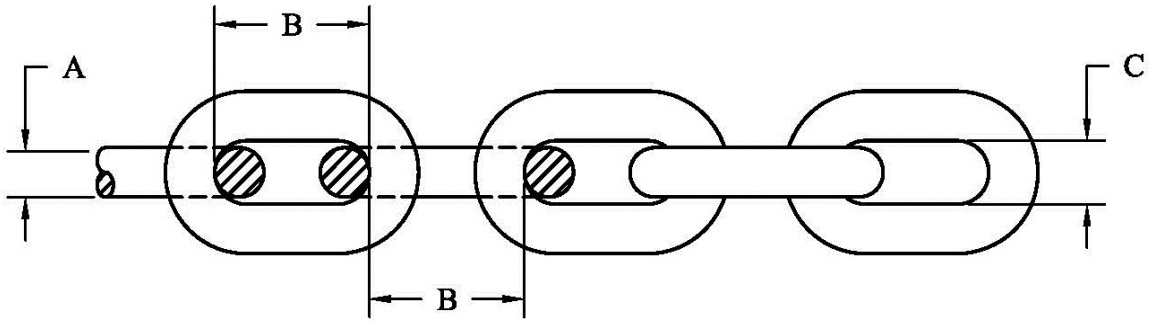


FIGURE 1-Type I, class 1, grade 50 chain.

TABLE I. Type I, class 1, grade 50 chain.

Trade Size (nom) (inch)	PIN size(1)	A (inch)	B (max) (inch)	C (min) (inch)	Working Load (lbf)
3/16	03	0.19	0.58	0.25	1,100
1/4	04	0.28	0.90	0.38	2,200
5/16	05	0.32	1.04	0.43	2,700
3/8	06	0.39	1.26	0.51	4,400
1/2	08	0.51	1.64	0.69	7,300
5/8	10	0.63	2.02	0.81	11,000

(1) PIN size is Trade size in 1/16 inch

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3.3.2.2 Type I, class 2, regular-link chain. The welded regular-link chain shall be similar to figure 2 and shall be in accordance with the dimensions and working loads shown in table II.

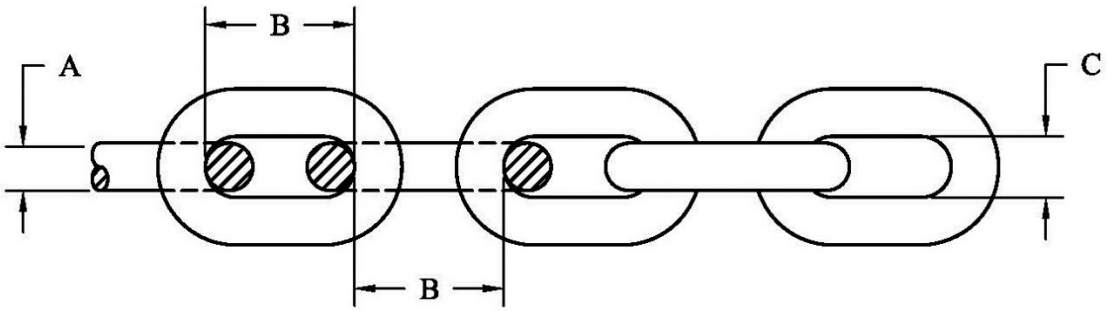


FIGURE 2. Type I, class 2, regular-link chain.

TABLE II. Type I, class 2, regular-link chain.

Trade Size (nom) (inch)	PIN size(1)	A (inch)	B (max) (inch)	C (min) (inch)	Working Load (lbf)
1/4	04	0.28	1.24	0.34	2,000
5/16	05	0.32	1.29	0.38	2,400
3/8	06	0.39	1.38	0.55	3,550
1/2	08	0.51	1.79	0.62	6,500

(1) PIN size is Trade size in 1/16 inch

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3.3.2.3 Type I, class 3, long-link chains. The welded long-link chain shall be similar to figure 3 and shall be in accordance with the dimensions and working loads shown in table III.

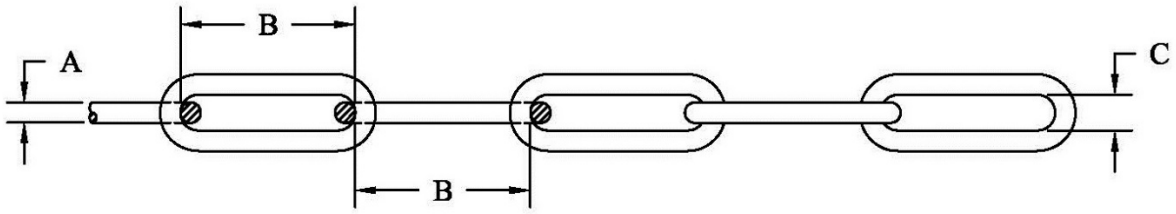


FIGURE 3. Type I, class 3, long-link chains.

TABLE III. Type I, class 3, long-link chains.

Trade Size (nom) (inch)	PIN size(1)	A (inch)	B (nom) (inch)	C (nom) (inch)	Working Load (lbf)
5/64	5X	0.08	0.48	0.14	180
3/32	3S	0.10	0.69	0.16	300
1/8	02	0.12	1.03	0.20	380
5/32	5S	0.15	1.26	0.35	500
3/16	03	0.20	1.42	0.39	900
1/4	04	0.24	1.64	0.47	1,200

(1) PIN size is Trade size in 1/16 inch except 5X which is 5/64 inch, and 3S and 5S which are 3/32 inch and 5/32 inch respectively

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3.3.2.4 Type I, class 5, twist-link chain. The welded twist link chain shall be similar to figure 4 and shall be in accordance with the dimensions and working loads shown in table IV.

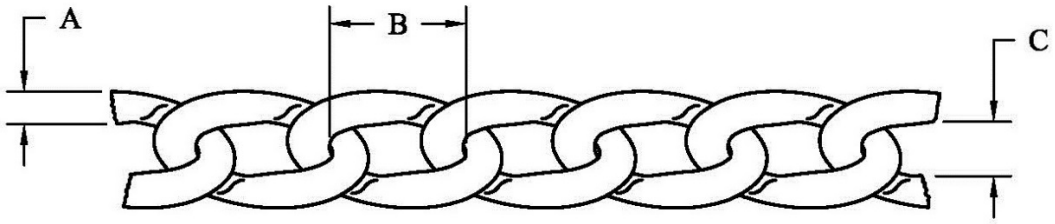


FIGURE 4. Type I, class 5, twist-link chain.

TABLE IV. Type I, class 5, twist-link chain.

Trade Size (nom) (inch)	PIN size[1]	A (inch)	B (nom) (inch)	C (nom) (inch)	Working Load (lbf)
5/64	05	0.07	0.30	0.13	150
3/32	06	0.09	0.31	0.16	190
1/8	08	0.11	0.48	0.22	280
9/64	09	0.14	0.53	0.24	400
5/32	10	0.15	0.59	0.26	500

(1) PIN size is Trade size in 1/64 inch

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3.4 Detail requirements - attachments. The figures herein are descriptive, not restrictive, and are not intended to preclude the purchase of attachments otherwise in accordance with this specification. Except as restricted by specified dimensions, actual design of all attachments is at the manufacturer's option.

3.4.1 General. The dimensions, physical requirements and other detailed data shall be as shown in the figures and respective tables.

3.4.1.1 Fabrication. All attachments and attachment components shall be forged, formed and welded, formed or machined as applicable unless otherwise noted. No attachment or attachment component shall be cast.

3.4.1.2 Working load limit. The working load limit (also shown as working load or WLL) is the maximum load that the attachment is rated to carry in non-shock loading conditions.

3.4.1.3 Proof and breaking loads. All attachments shall be proof and break tested as required at the working load limit factor specified for that item. Fixture shall be in accordance with 4.3.2.1.3.

3.4.1.4 Proof test. Attachments withstanding the proof load without loss of integrity or detrimental deformation shall have passed the proof test. Except as noted, detrimental deformation shall be considered to be when the deformation of an attachment or feature of an attachment such as a turnbuckle eye exceeds 2% or when the attachment cannot be normally operated. Marks due to attachment to attachment or attachment to fixture contact shall not be considered as signs of loss of integrity or deformation. Type VI rings, type XIII pear-shaped links and type XVI master links are exempt from the deformation requirement.

3.4.1.5 Ductility. All attachments except type XIV mechanical coupling links shall show ductility of not less than 15% or as otherwise specified after break testing. Attachments are not required to be tested to failure provided the minimum breaking load is exceeded.

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3.4.2 Type II, connecting link. The connecting links shall be forged and be similar to figure 5 and shall be in accordance with the dimensions and working loads shown in table V.

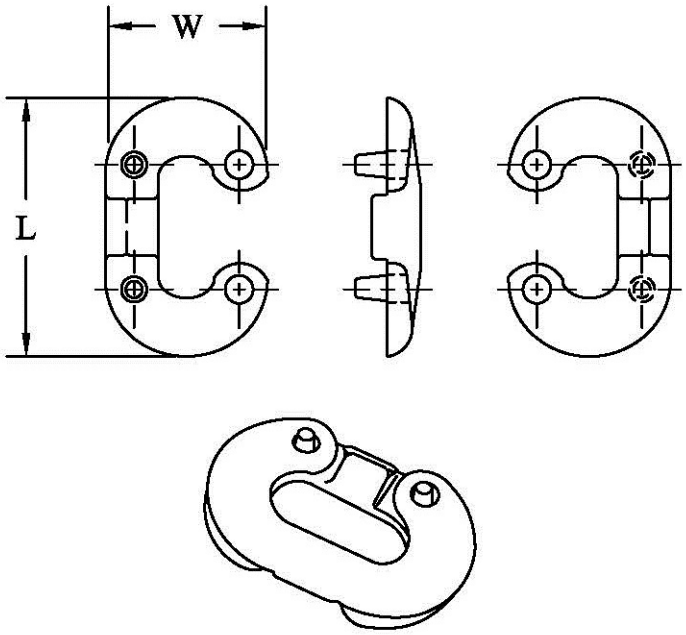


FIGURE 5. Type II, connecting link.

TABLE V. Type II, connecting link.

Trade Size (nominal inch)	PIN size(1)	W (approx.) (inch)	L (approx.) (inch)	Working Load (lbf)
1/4	04	1.00	1.50	1,200

(1) PIN size is Trade Size in 1/16 inch

3.4.2.1 Interchangeability. Any randomly selected connecting link halves shall be interchangeable without applying force.

3.4.2.2 Rivet size. The rivets shall be of sufficient diameter and length to completely fill the corresponding rivet hole when peened.

3.4.2.3 Proof and break load factor. For connecting links the proof load shall be a minimum of 2 times the working load limit. The break load shall be a minimum of 4 times the working load limit.

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3.4.3 Type IV, Shackles.

3.4.3.1 General. The shackles shall be in accordance with the dimensions and working loads shown in the respective figures and tables. Shackles shall be forged from stainless steel. The body and pin shall be of the same alloy.

3.4.3.1.1 Threads. The screw-pin and bolt shackles shall be threaded in accordance with ASME B1.1 UNC/UNCR threads no looser than class 1 fit unless otherwise specified (see 6.2).

3.4.3.1.2 Working load limit. The working load limits (WLL) of shackles covered by this specification are shown in table VI.

TABLE VI. Type IV shackle mechanical properties.

Trade size (nom) (inches)	PIN Size(1)	Working load limit (min) (lbf)
1/4	04	1,000
5/16	05	1,300
3/8	06	1,500
7/16	07	2,000
1/2	08	3,000
5/8	10	4,000
3/4	12	6,000
7/8	14	8,000
1	16	10,000
1-1/4	20	14,000

(1) PIN size is Trade size in 1/16 inch

3.4.3.1.3 Proof loads. Shackles shall withstand a proof load of 2.2 times (2.2x) the Working Load Limit (WLL) shown in table VI when tested in accordance with 4.4.2.3 without developing surface rupture or defects that would interfere with serviceability or prevent disassembly of the pin. After proof loading, screw-pin shackles shall be disassembled by hand after the first turn.

3.4.3.1.4 Breaking loads. Shackles shall exceed a breaking load of 5x the WLL requirements shown in table VI when test in accordance with 4.4.2.4.

3.4.3.1.5 Ductility. Shackles shall be sufficiently ductile so that, when fractured, the fractured member shall show a permanent distortion before breaking. If the pin fractures, it shall show a permanent bend of not less than 20 degrees. If the body fractures, either the length from bearing point to bearing point (dimension L) or the bow width (dimension B) must deform 15% or more from the original dimension. Shackles that are not broken during the Breaking test (see 4.4.2.4) shall meet the ductility requirements at the highest applied load.

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3.4.3.1.6 Marking. Each shackle body shall be permanently and legibly marked in raised or stamped letters on the side of the shackle bow with the identifying manufacturer's name, logo or trademark, shackle size, alloy designation and the working load limit. The pins shall be marked with the manufacturer's logo or trademark and the alloy. The WLL may be in pounds or tons at the manufacturer's option. Shackle markings shall be raised or stamped letters or figures of the maximum practical height permitted by the size of the shackle component being marked, but not to exceed 0.75 inch in height by 0.125inch in relief. Stamping dies shall be of the round bottom, low stress type. Marking location shall not interfere with the serviceability of the shackle assembly.

3.4.3.2 Form and dimensions. The form of the respective shackle types and classes shall be similar to that shown in figures 6 and 7 (see 6.2). Except for reference dimensions, the dimensions of shackle bodies and component pins and bolts shall conform to the dimensional requirements specified in the applicable tables. Reference dimensions are for information only.

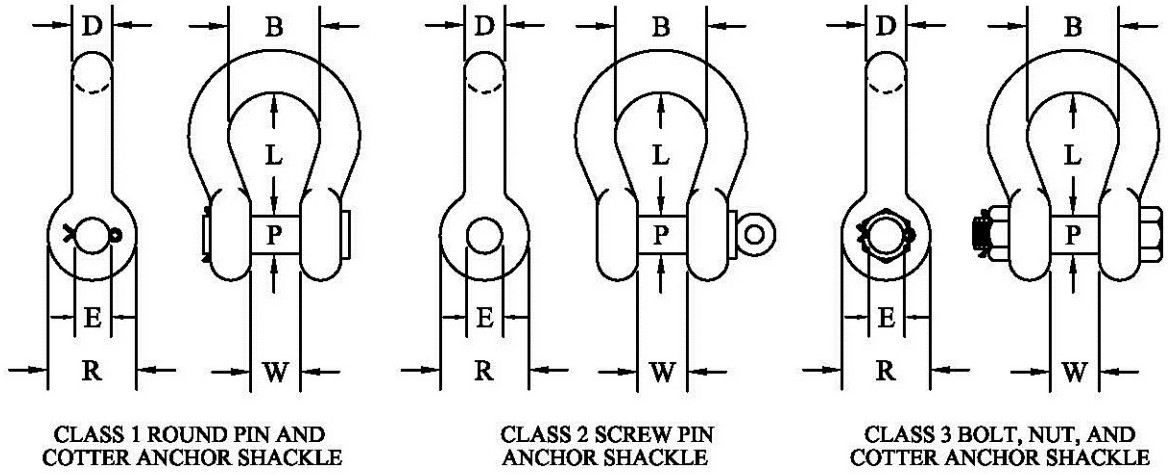


FIGURE 6. Type IVA, anchor shackles.

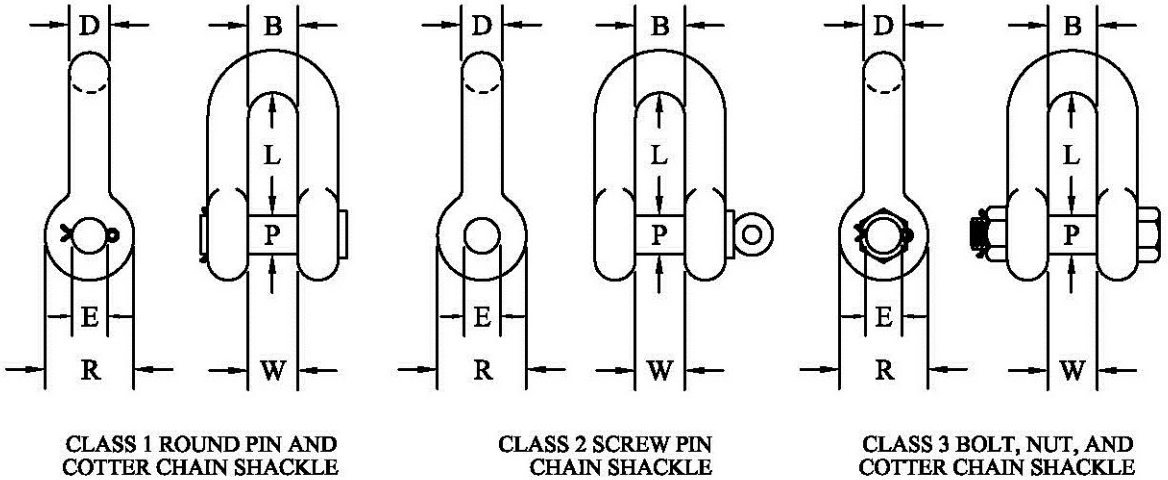


FIGURE 7. Type IVB, chain shackles.

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TABLE VII. Type IVA, anchor shackles, all classes.

Trade size (nom) (inches)	PIN Size(1)	Dimension D (min) (inches)	Diameter pin P (min) (inches)	Diameter unthreaded eye E(2) (ref) (inches)	Width between eyes W (inches)	Length inside L (inches)	Width bow B (min) (inches)	Diameter outside eye R (max) (inches)
1/4	04	.24	.30	.40	.47±0.13	1.13±.06	.75	.88
5/16	05	.30	.36	.47	.53±0.13	1.25±.06	.81	1.00
3/8	06	.36	.43	.53	.66±0.13	1.44±.13	.93	1.13
7/16	07	.42	.49	.59	.75±0.13	1.69±.13	1.06	1.25
1/2	08	.48	.61	.72	.81±0.13	1.88±.13	1.18	1.38
5/8	10	.59	.73	.84	1.06±0.13	2.38±.13	1.50	1.88
3/4	12	.71	.85	.97	1.25±0.13	2.82±.25	1.75	2.13
7/8	14	.83	.98	1.09	1.44±0.13	3.31±.25	2.00	2.38
1	16	.95	1.10	1.22	1.69±0.13	3.75±.25	2.31	2.63
1-1/4	20	1.19	1.34	1.47	2.03±0.13	4.69±.25	2.87	3.25

(1) PIN size is Trade size in 1/16 inch
(2) Threaded eye on class 2 shackles shall meet thread class requirements

TABLE VIII. Type IVB, chain shackles, all classes.

Trade size (nom) (inches)	PIN size	Dimension D (min) (inches)	Diameter pin P (min) (inches)	Diameter unthreaded eye E(2) (ref) (inches)	Width between eyes W (inches)	Length inside L (inches)	Diameter outside eye R (max) (inches)
1/4	04	.24	.30	.40	.47±.13	.88±.06	.88
5/16	05	.30	.37	.47	.53±.13	1.03±.06	1.00
3/8	06	.36	.43	.53	.66±.13	1.13±.13	1.25
7/16	07	.42	.49	.59	.75±.13	1.44±.13	1.25
1/2	08	.48	.61	.72	0.81±.13	1.63±.13	1.38
5/8	10	.59	.73	.84	1.06±.13	2.00±.13	1.88
3/4	12	.71	.85	.97	1.25±.13	2.38±.25	2.13
7/8	14	.83	.98	1.09	1.44±.13	2.81±.25	2.38
1	16	.95	1.10	1.22	1.69±.13	3.19±.25	2.63

(1) PIN size is Trade size in 1/16 inch
(2) Threaded eye on class 2 shackles shall meet thread class requirements

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3.4.4 Type V, Hooks.

3.4.4.1 General. The hooks shall be in accordance with the dimensions and working loads shown in the respective figures and tables. Style 1 and 2 hook connections shall be compatible type 1 chain of the equivalent size. Style 5 hook swivel connections shall be compatible with type 1 chain of equivalent capacity.

3.4.4.1.1 Construction. All hook styles shall be of forged construction.

3.4.4.1.2 Latches. Style 1 and style 5 hooks will be capable of accepting a securing latch mechanism. The mechanism shall be easily installable/removable. The latch mechanism shall only be supplied if specified in the procurement documents (see 6.2) and may be installed or provided in the hook packaging at the manufacturer's option.

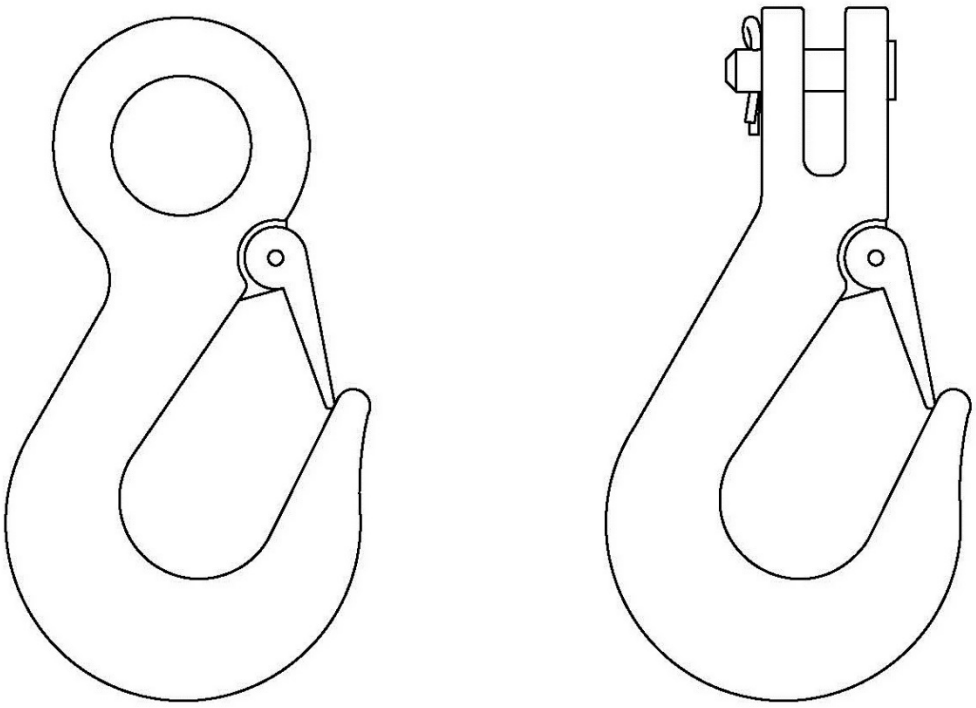
3.4.4.1.3 Proof and break load factor. For all hook styles the proof load shall be a minimum of 2 times the working load limit. The break load shall be a minimum of 4 times the working load limit.

3.4.4.1.3.1 Proof test. Hooks shall withstand the proof test with no more than 1% permanent opening of the hook at the throat. All other dimensions shall not exceed 2% deformation.

3.4.4.1.3.2 Breaking load and ductility tests. Samples subjected to the breaking load strength test shall exhibit permanent deformation in the throat opening after failure or at the maximum applied load.

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3.4.4.2 Type V, style 1 slip hook. The slip hook is used to attach a chain to a load. Slip hooks shall be similar to figure 8 and shall be in accordance with the working loads shown in table IX.



(a) Eye style (style 1E) (b) Round pin clevis style (style 1C)

FIGURE 8. Basic hook design – style 1 slip hook.

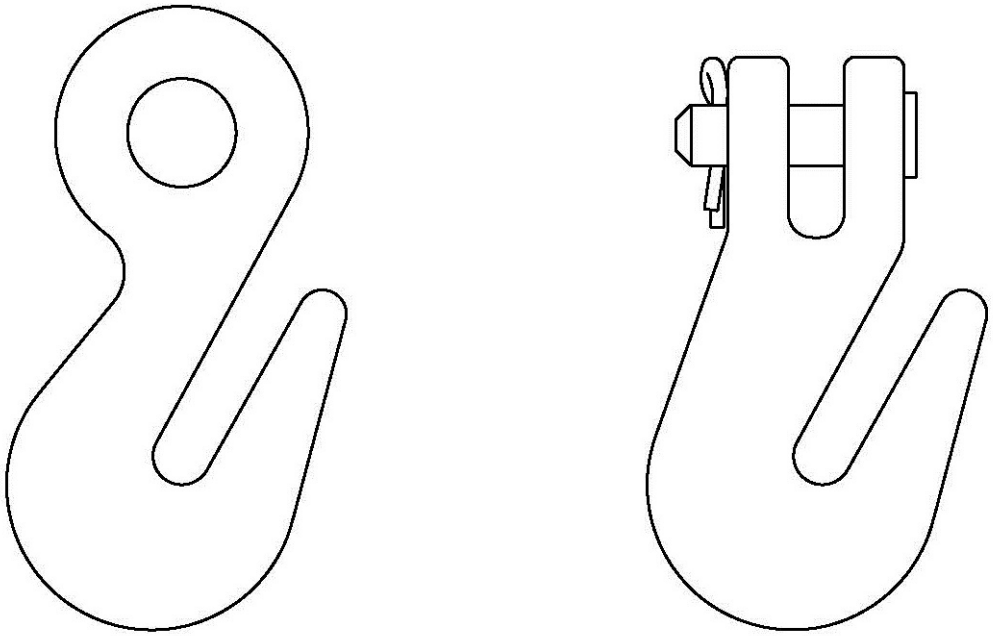
TABLE IX. Type V, style 1E, slip hook.

Size (nom) (inches)	PIN size(1)	Working load limit (lbf)
1/4	04	900
5/16	05	1,250
3/8	06	2,000
1/2	08	3,000
5/8	10	8,000
3/4	12	12,000
1	16	20,000

(1) PIN size is size (nom) in 1/16 inch.

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3.4.4.3 Type V, style 2 grab hook. The grab hook is used to attach a chain to a specific link of a chain. Grab hooks shall be similar to that shown on figures 9 and shall be in accordance with the working loads shown in table X.



(a) Eye attachment (style 2E) (b) Round pin clevis attachment (style 2C)

Figure 9. Basic hook design – style 2 grab hook.

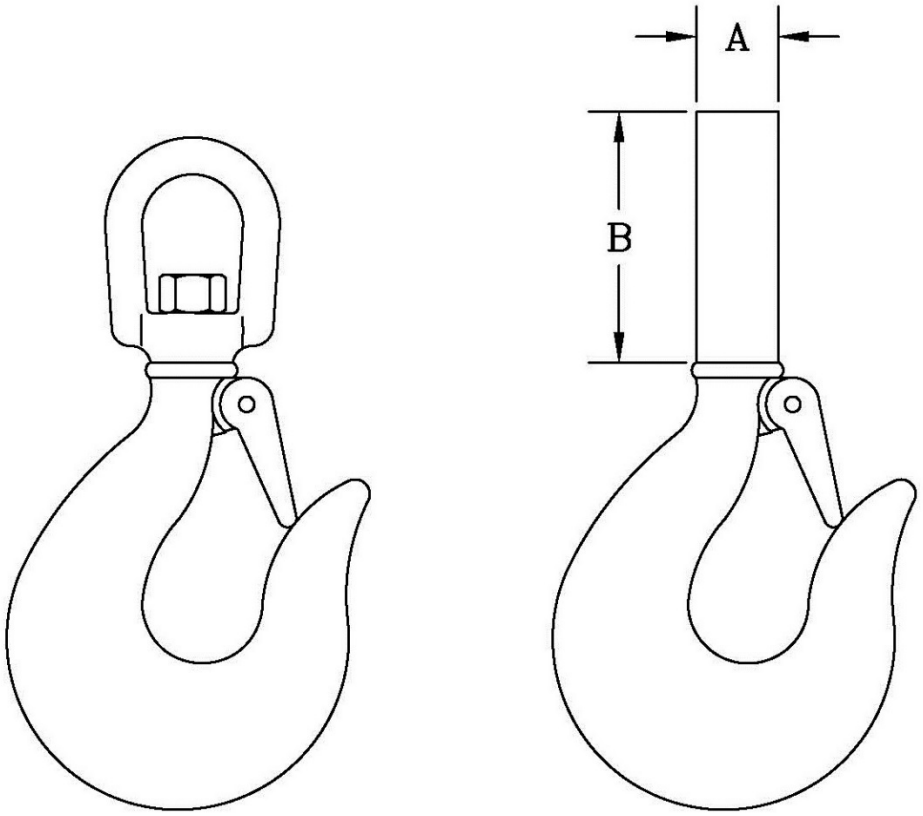
TABLE X. Type V, style 2 grab hook.

Size (nom) (inch)	PIN size number (1)	Working load limit (lbf)
1/4	04	1,500
5/16	05	2,000
3/8	06	2,500
1/2	08	3,600

(1) PIN size is size (nom) in 1/16 inch

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3.4.4.4 Type V, style 5 hoist hook. The hoist hook is typically used with wire rope, not chain, to attach to a load. Hoist hooks shall similar to that shown on figures 10 and shall be in accordance with the working loads and dimensions shown in table XI.



(a) Swivel attachment (style 5S) (b) Shank attachment (style 5T and 5U)

FIGURE 10. Basic hook design – style 5 hoist hook.

TABLE XI. Type V, style 5, hoist hook.

Hook Size	PIN size number(1)	Working load limit (lbf)	A (inch) (min)(2) (5T and 5U only)	B (inch) (min)(2) (5T and 5U only)
.5 ton	050	1,000	.50	2.00
.75 ton	075	1,500	.63	2.25
1 ton	100	2,000	.75	2.50
1.25 ton	125	2,500	.88	2.75

(1) PIN size is WLL in 1/100 tons
(2) For 5T hooks, the thread size shall be “A”.

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3.4.4.4.1 Shank style. Shank hooks shall be provided with unthreaded shanks (style 5U) or threaded shanks (style 5T) as specified in the contract (see 6.2). The threaded shank shall be ASME B1.1 UNC/UNCR thread of dimension "A" size given in table XI.

3.4.5 Type VI, rings. The rings shall be forged or formed and welded and be similar to that shown on figures 11 and shall be in accordance with the dimensions and working loads shown in table XII.

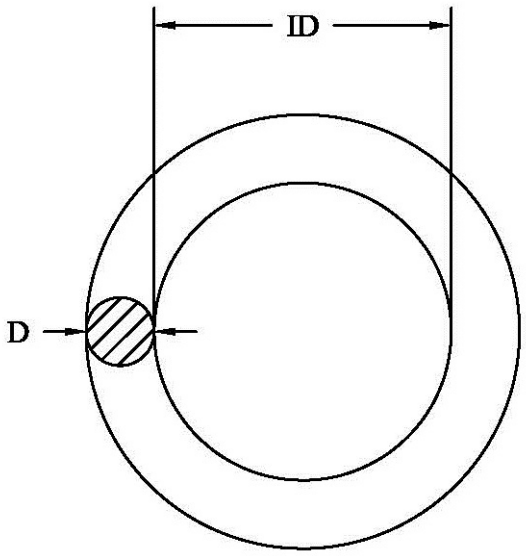


FIGURE 11. Type VI, ring.

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TABLE XII. Type VI, ring.

Diameter stock (D) (nom) (inches)	Inside Diameter (ID) (nom) (inches)	PIN size(1)	Working load (lbf)
1/8	3/4	0206	125
1/8	1	0208	125
5/32	1	5S08	300
5/32	1-1/2	5S12	250
3/16	1	0308	500
3/16	1-1/4	0310	300
3/16	1-1/2	0312	300
3/16	2	0316	250
1/4	1	0408	600
1/4	1-1/4	0410	500
1/4	1-1/2	0412	400
1/4	2	0416	350
1/4	3	0424	250
5/16	1-1/2	0512	750
5/16	1-3/4	0514	700
5/16	2	0516	600
5/16	2-1/2	0520	500
3/8	2	0616	1,100
3/8	2-1/2	0620	900
3/8	3-1/4	0626	800
1/2	2-3/8	0819	1,400
1/2	2-1/2	0820	1,350
1/2	3-1/4	0826	1,000
1/2	4	0832	800

(1) PIN size is (D) in 1/16 inch in 2 digits, except 5S is D = 5/32 inch, followed by inside diameter in 1/8 inch

3.4.5.1 Proof and break load factor. For rings the proof load shall be a minimum of 2 times the working load limit. The break load shall be a minimum of 4 times the working load limit.

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3.4.6 Type VII – swivels.

3.4.6.1 General. The dimensions, physical requirements, weights, and other detailed data shall be as shown in the figures and respective tables.

3.4.6.1.1 Proof and break load factor. For all swivels the proof load shall be a minimum of 2.2 times the working load limit. The break load shall be a minimum of 5 times the working load limit.

3.4.6.1.2 Closure pins. For class 3 and 4 swivels the pin may be of clevis pin and cotter pin (class suffix C) or bolt, nut and cotter pin (class suffix B) at the option of the manufacturer unless a specific style is specified in the order (see 6.2). The PIN shall be for the style supplied.

3.4.6.2 Type VII, class 1, chain swivel. The chain swivels shall be similar to that shown on figures 12 and shall be in accordance with the dimensions, working loads and weights as shown in table XIII.

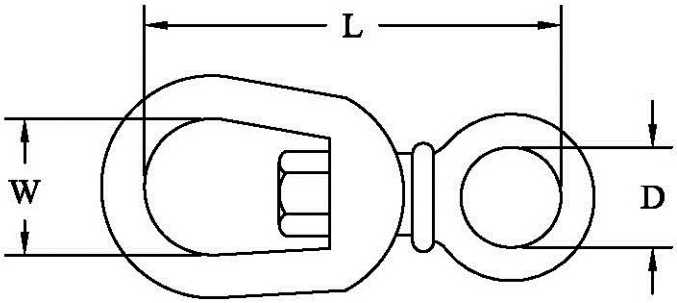


FIGURE 12. Type VII, class 1, chain swivel.

TABLE XIII. Type VII, class 1, chain swivel.

Trade size (nom) (inches)	Length L (min) (inches)	Diameter D (min) (inches)	Width W (nom) (inches)	PIN Size(1)	Working load (lbf)
1/4	2.25	.43	.75	08	900
5/16	2.72	.50	1.00	10	1,200
3/8	3.44	.75	1.25	12	2,600
1/2	4.25	.87	1.50	16	4,400
5/8	5.13	1.06	1.75	20	6,000

(1) PIN size is Trade size in 1/32 inch

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3.4.6.3 Type VII, class 2, eye and eye swivel. The eye and eye swivels shall be similar to that shown on figures 13 and shall be in accordance with the dimensions and working loads shown in table XIV.

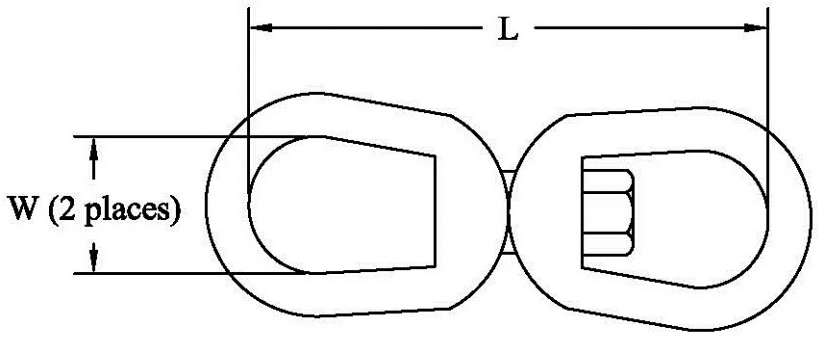


FIGURE 13. Type VII, class 2, eye and eye swivel.

TABLE XIV. Type VII, class 2, eye and eye swivel.

Trade size (nom) (inches)	Length L (min) (inches)	Width W (nom) (inches)	PIN size(1)	Working load (lbf)
1/4	2.88	.75	08	900
3/8	4.25	1.25	12	2,600
1/2	5.42	1.50	16	4,400
5/8	6.55	1.75	20	6,000

(1) PIN size is Trade size in 1/32 inch

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3.4.6.4 Type VII, class 3B and class 3C, eye and jaw swivel. The eye and jaw swivels shall be similar to that shown on figures 14 (class 3B is shown) and shall be in accordance with the dimensions and working loads as shown in table XV. Class 3B shall be a bolt, nut and cotter pin closure while class 3C shall be a clevis pin and cotter pin closure, otherwise the classes are the same.

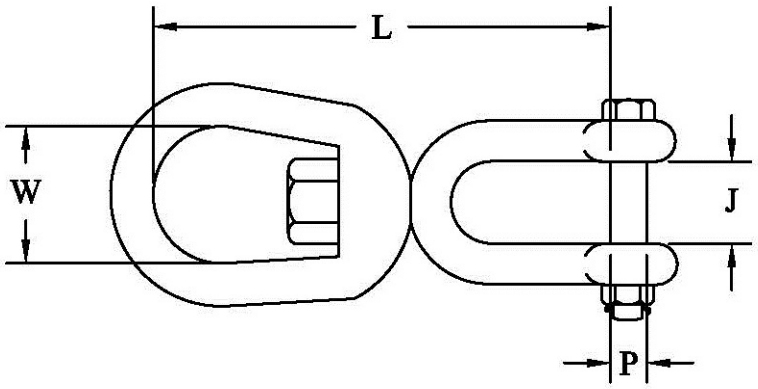


FIGURE 14. Type VII, class 3, eye and jaw swivel.
(Only class 3B is shown, class 3C is similar)

TABLE XV. Type VII, class 3, eye and jaw swivel.

Trade size (nom) (inches)	Length L (min) (inches)	Width J (min) (inches)	Width W (nom) (inches)	Diameter Pin P (nom) (inches)	PIN Size(1)	Working load (lbf)
1/4	2.43	.46	.75	.25	08	900
5/16	2.81	.50	1.00	.31	10	1,600
3/8	3.50	.62	1.25	.37	12	2,600
1/2	4.50	.75	1.50	.50	16	4,400
5/8	5.31	.93	1.75	.62	20	6,000

(1) PIN size is Trade size in 1/32 inch

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3.4.6.5 Type VII, class 4B and class 4C, jaw and jaw swivel. The jaw and jaw swivels shall be similar to that shown on figures 15 (class 4B is shown) and shall be in accordance with the dimensions, working loads and weights as shown in table XVI. Class 4B shall be bolt, nut and cotter pin closures while class 4C shall be clevis pin and cotter pin closures, otherwise the classes are the same. Both closures shall be the same.

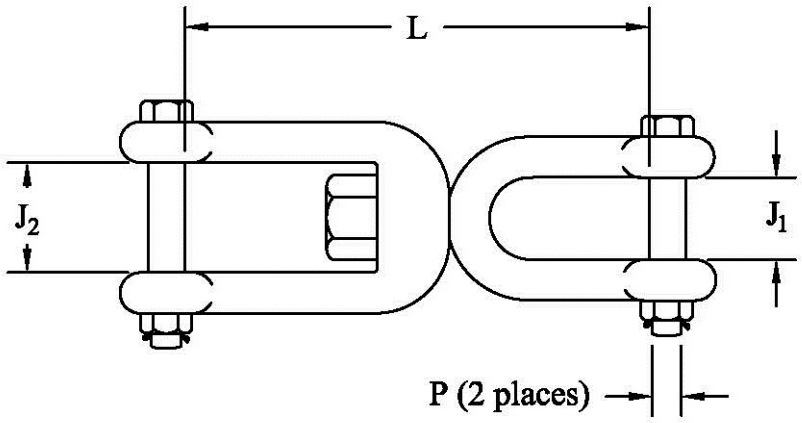


FIGURE 15. Type VII, class 4, jaw and jaw swivel.
(Only class 4B is shown, class 4C is similar)

TABLE XVI. Type VII, class 4, jaw and jaw swivel.

Trade size (nom) (inches)	Length L (min) (inches)	Width J_1 (min) (inches)	Width J_2 (min) (inches)	Diameter Pin P (nom) (inches)	PIN Size(1)	Working load (lbf)
3/16	2.20	.44	.53	0.18	06	600
1/4	2.69	.46	.60	0.25	08	900
5/16	2.89	.50	.66	0.31	10	1,600
3/8	3.64	.62	.86	0.37	12	3,400
5/8	5.11	.93	1.23	0.62	20	6,000

(1) PIN size is Trade size in 1/32 inch

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3.4.7 Type XIII – pear-shaped link. The pear links shall be forged or formed and welded and be similar to figure 16, and shall be in accordance with the dimensions and working loads shown in table XVII.

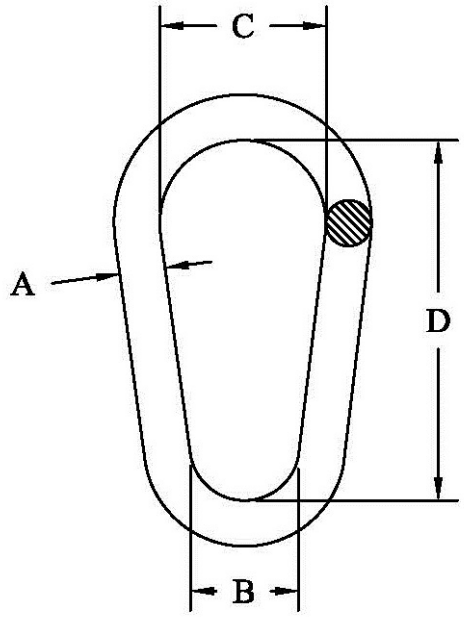


FIGURE 16. Type XIII pear-shaped link.

TABLE XVII. Type XIII, pear-shaped link.

Diameter stock (A) (nom) (inches)	Width (B) (nom) (inches)	Width (C) (nom) (inches)	Length (D) (nom) (inches)	PIN Size(1)	Working load (lbf)
5/8	1.25	2.50	3.75	20	2,000
1	2.00	3.50	6.00	32	3,750

(1) PIN size is (A) in 1/32 inch

3.4.7.1 Proof and break load factor. For pear-shaped links the proof load shall be a minimum of 2 times the working load limit. The break load shall be a minimum of 4 times the working load limit.

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3.4.8 Type XIV – mechanical coupling link. The mechanical coupling links shall be similar to figure 17 and shall be in accordance with the dimensions and working loads shown in table XVIII.

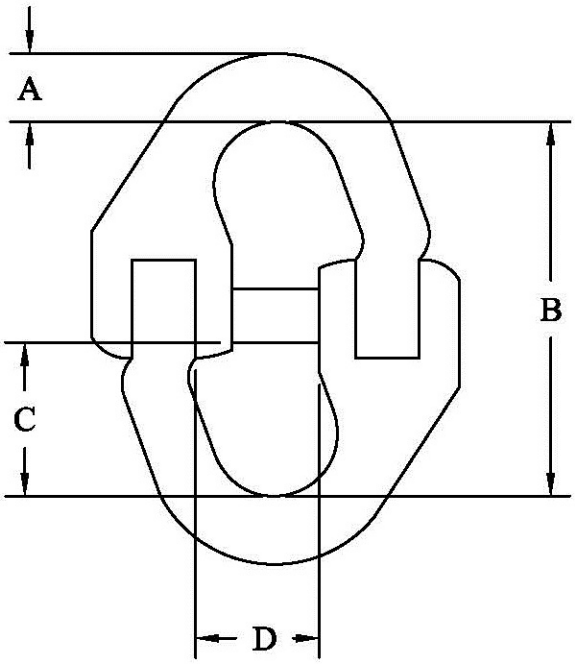


FIGURE 17. Type XIV mechanical coupling link.

TABLE XVIII. Type XIV mechanical coupling link.

Trade size (inches)	A (inches)		B (inches)		C (inches)		D (inches)		PIN Size(1)	Working load (lbf)
	min	max	min	max	min	max	min	max		
3/16	.25	.35	1.34	1.54	.46	.66	.38	.58	03	1,100
1/4	.30	.40	1.84	2.04	.65	.85	.47	.67	04	2,100
5/16	.36	.49	2.13	2.81	.83	.89	.68	1.00	05	2,700
3/8	.43	.59	2.33	3.06	.83	1.18	.81	1.13	06	4,400
1/2	.58	.78	3.16	3.86	1.20	1.48	1.00	1.31	08	7,300
5/8	.71	.91	3.80	4.60	1.37	1.74	1.25	1.60	10	11,000

(1) PIN size is Trade size in 1/16 inch

3.4.8.1 Proof and break load factor. For mechanical coupling links the proof load shall be a minimum of 2 times the working load limit. The break load shall be a minimum of 4 times the working load limit.

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3.4.9 Type XVI – master link. The master links shall be formed or forged and shall be similar to figure 18 and shall be of sufficient size to meet the WLL shown in table XIX.

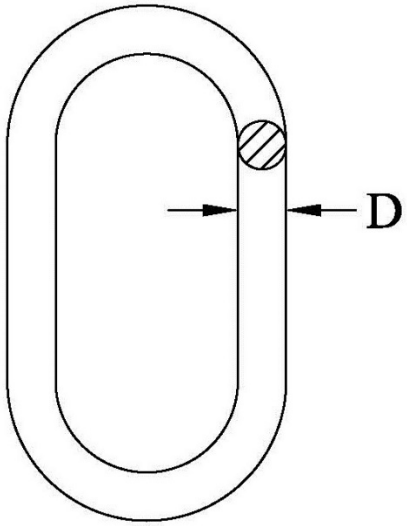


FIGURE 18. Type XVI master link.

TABLE XIX. Type XVI master link.

Trade size (D) (nom) (inches)	PIN Size(1)	Working load – single leg (min) (lbf)
3/8	06	1,500
1/2	08	2,000
5/8	10	2,800
3/4	12	3,750
1	16	8,000
1-1/4	20	11,800

(1) PIN size is Trade size in 1/16 inch

3.4.9.1 Proof and break load factor for single leg assemblies. For individual master links and single leg assemblies the proof load shall be a minimum of 2 times the working load limit. The break load shall be a minimum of 4 times the working load limit.

3.4.9.2 Proof testing for multi-leg assemblies. Multi-leg assemblies shall be tested per ASTM A952/A952M using the values in table XIX.

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3.4.10 Type XVII – Links, loops and D-rings.

3.4.10.1 General. The dimensions and working loads shall be as shown in the respective sections.

3.4.10.1.1 Proof and break load factor. For type XVII items the proof load shall be a minimum of 2 times the working load limit. The break load shall be a minimum of 4 times the working load limit.

3.4.10.2 Type XVII, class 1, rectangular link. The rectangular links shall be similar to figure 19, and shall be in accordance with the dimensions and working loads shown in table XX.

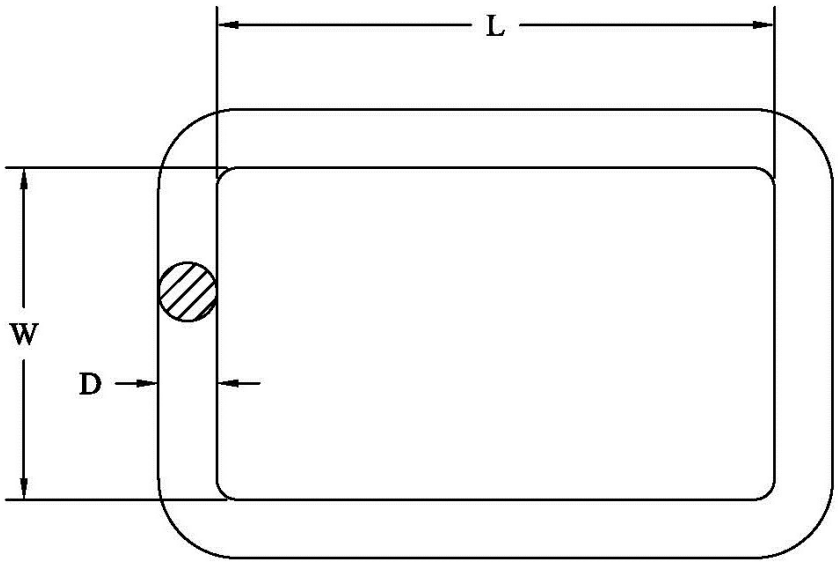


FIGURE 19. Type XVII, class 1, rectangular link.

TABLE XX. Type XVII, class 1, rectangular link.

Diameter stock (D) (nom) (inches)	Width (W) (min) (inches)	Length (L) (min) (inches)	PIN size(1)	Working load (lbf)
3/16	0.50	1.00	0304	600
3/16	0.50	2.00	0308	550
1/4	1.00	2.00	0408	800
5/16	1.50	2.50	0506	1,200
5/16	1.50	3.00	0512	1,100

(1) PIN size is diameter (D) in 1/16 inch (2 digits) plus length (L) in 1/4 inch (2 digits)

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3.4.10.3 Type XVII, class 2, triangle loop. The triangle loops shall be forged or formed and welded and be similar to that shown on figures 20 and shall be in accordance with the dimensions and working loads shown in table XXI.

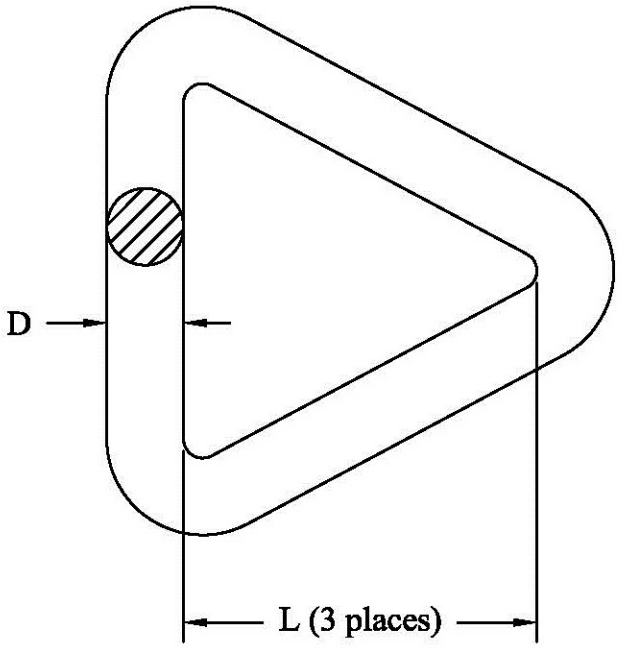


FIGURE 20. Type XVII, class 2, triangle loop.

TABLE XXI. Type XVII, class 2, triangle loop.

Diameter stock (D) (nom) (inches)	Length (nom) (inches)	PIN size(1)	Working load (lbf)
3/16	1.00	03	500
1/4	2.00	04	750

(1) PIN size is (D) in 1/16 inch

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3.4.10.4 Type XVII, class 3, D-ring. The D-rings shall be forged or formed and welded and be similar to that shown on figures 21 and shall be in accordance with the dimensions and working loads shown in table XXII.

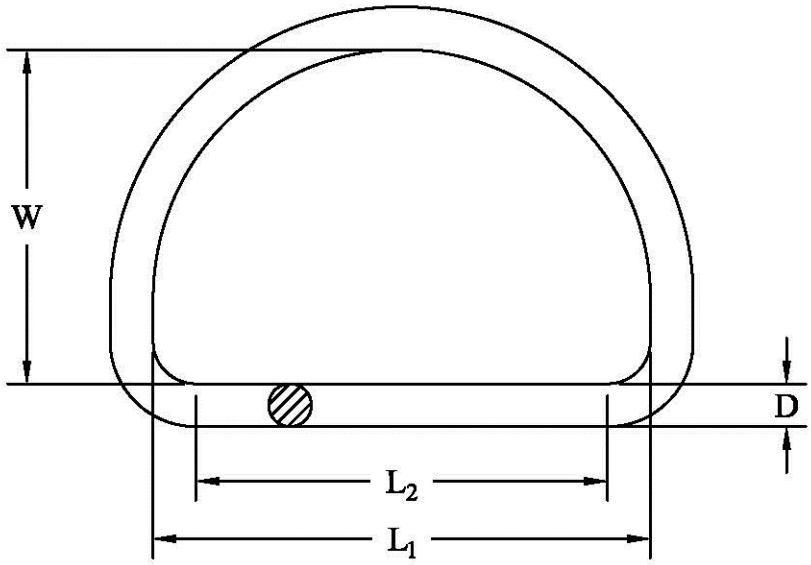


FIGURE 21. Type XVII, class 3, D-ring.

TABLE XXII. Type XVII, class 3, D-ring.

Diameter stock (D) (nom) (inches)	Length L_1 (nom) (inches)	Length L_2 (nom) (inches)	Width (W) (nom) (inches)	PIN Size(1)	Working load (lbf)
1/8	1.50	1.25	0.88	021	100
3/16	1.75	1.38	1.06	031	300
3/16	2.25	1.88	1.31	032	300
1/4	2.00	1.50	1.25	041	500
1/4	3.00	2.50	1.75	043	500
5/16	3.25	2.62	1.94	051	800
5/16	3.75	3.13	2.19	052	800
5/16	4.25	2.44	2.44	053	800

(1) PIN size is (D) in 1/16 inch (2 digits) plus a sequential number

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3.4.10.5 Type XVII, class 4, heavy-duty D-ring. The heavy-duty D-rings shall be forged and similar to that shown on figures 22 and shall be in accordance with the dimensions and working loads shown in table XXIII.

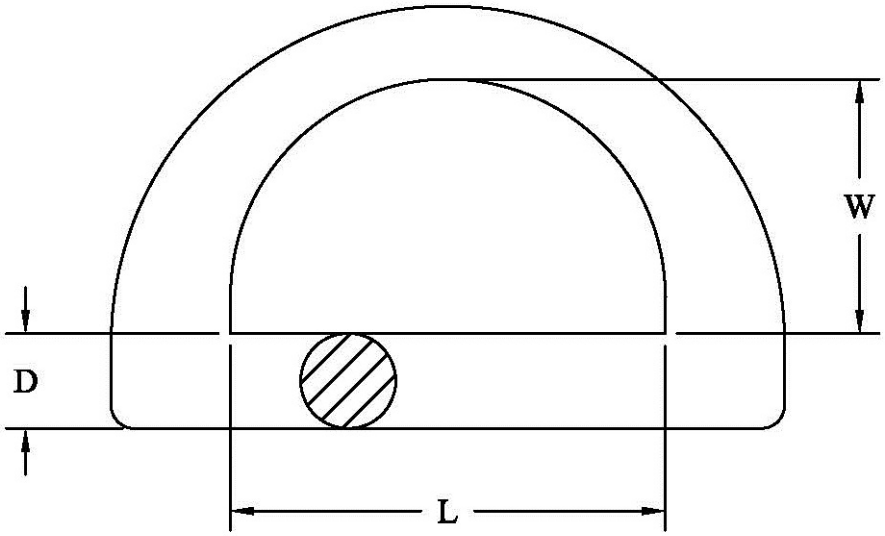


FIGURE 22. Type XVII, class 4, heavy-duty D-ring.

TABLE XXIII. Type XVII, class 4, heavy-duty D-ring.

Diameter stock (D) (nom) (inches)	Length (L) (nom) (inches)	Width (W) (nom) (inches)	PIN Size(1)	Working load (lbf)
1/4	1.00	0.68	04	1,200
5/16	1.25	0.87	05	2,000

(1) PIN size is (D) in 1/16 inch

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3.4.11 Type XVIII - turnbuckles.

3.4.11.1 General. The dimensions, physical requirements and other detailed data shall be as shown in the respective figures and tables. Jaw and eye dimensions not specified in the tables shall be in accordance with ASTM F1145.

3.4.11.1.1 Fabrication method. Turnbuckle bodies and jaw and eye components shall be forged. Jaw component bolts and nuts, and jam nuts shall be forged or machined at the manufacturer's option.

3.4.11.1.2 Turnbuckle threads. Turnbuckle threads shall not be looser than ASME B1.1 UNC/UNCR class 1A/1B and equivalent to trade size A dimension.

3.4.11.1.3 Jam Nuts. Jam nuts may be included at the manufacturer's option but are not required unless option Y is specified in the procurement documents. PIN shall represent actual configuration supplied.

3.4.11.1.4 Proof and break load factor. For all turnbuckles the proof load shall be a minimum of 2.2 times the working load limit. The break load shall be a minimum of 5 times the working load limit.

3.4.11.1.5 Turnbuckle assemblies. All turnbuckle assemblies class 1 through 4 shall use the appropriate class 5 turnbuckle body.

3.4.11.1.6 Closure pins. For class 1 and 2 turnbuckles the pin may be of clevis pin and cotter pin (class suffix C) or bolt, nut and cotter pin (class suffix B) at the option of the manufacturer unless a specific style is specified in the order (see 6.2). The PIN shall be for the style supplied.

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3.4.11.2 Type XVIII, class 1B and class 1C, jaw and jaw turnbuckle. The jaw and jaw turnbuckles shall be similar to that shown on figure 23, and shall be accordance with the dimensions and working loads shown in table XXIV. Class 1B shall be a bolt, nut and cotter pin closure while class 1C shall be a clevis pin and cotter pin closure, otherwise the classes are the same. Both closures shall be the same

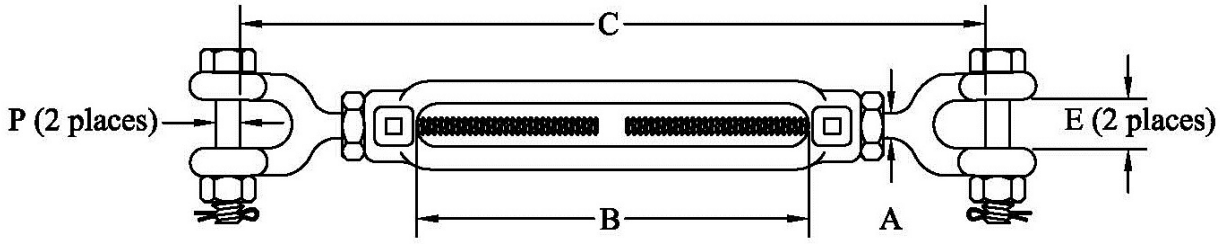


FIGURE 23. Type XVIII, class 1, jaw and jaw turnbuckle.
(Only class 1B is shown, class 1C is similar)

TABLE XXIV. Type XVIII, class 1, jaw and jaw turnbuckle.

Trade Size (Thread Size) A (nom) (inch)	Clear Opening B (min)(inch)	PIN size number(1)	Length C (max at full retraction)(2) (inch)	Pin dia. P (nom)(inch)	Width E (min)(inch)	Working load (lbf)
1/4	4.00	0404	7.50	0.25	0.44	500
5/16	4.50	0504	8.75	0.25	0.50	700
3/8	6.00	0606	10.31	0.31	0.56	1,040
1/2	6.00	0806	11.62	0.38	0.63	1,800
5/8	6.00	1006	13.00	0.50	0.75	2,700
3/4	6.00	1206	14.37	0.63	1.00	4,000
1	6.00	1606	17.12	0.88	1.25	7,600
1	12.00	1612	23.12	0.88	1.25	7,600
1-1/4	12.00	2012	25.50	1.13	1.88	12,000
1-1/2	12.00	2412	27.60	1.36	2.19	13,400

(1) PIN size is trade size (A) in 1/16 inch including leading zero if required and clear opening (B) in full inches including leading zero if required
(2) Dimension “C” is without jam nuts installed

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3.4.11.3 Type XVIII, class 2B and 2C, jaw and eye turnbuckle. The jaw and eye turnbuckles shall be similar to that shown on figure 24, and shall be accordance with the dimensions and working loads shown in table XXV. Class 2B shall be a bolt, nut and cotter pin closure while class 2C shall be a clevis pin and cotter pin closure, otherwise the classes are the same.

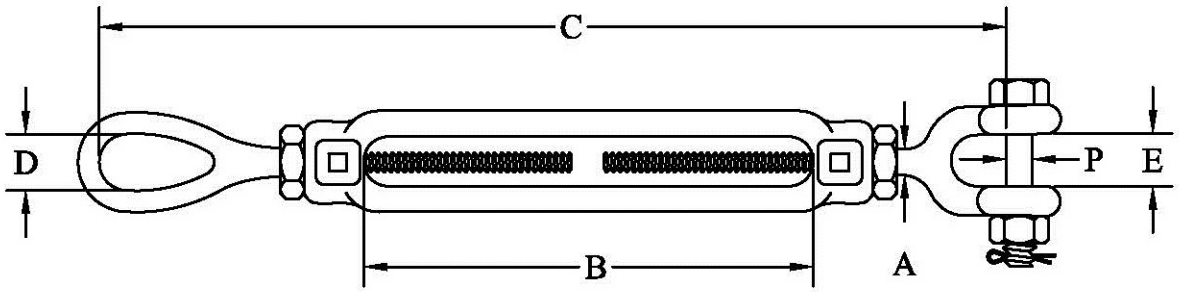


FIGURE 24. Type XVIII, class 2, jaw and eye turnbuckle.
(Only 2B is shown, class 2C is similar)

TABLE XXV. Type XVIII, class 2, jaw and eye turnbuckle.

Trade Size A (Thread Size) (nom) (inch)	Clear Opening B (min)(inch)	PIN size number(1)	Length C (max at full retraction)(2) (inch)	Pin dia. P (nom)(inch)	Jaw Width E (min)(inch)	Eye Width D (min)(inch)	Working load (lbf)
1/4	4.00	0404	7.92	0.25	0.44	0.33	500
5/16	4.50	0504	9.34	0.25	0.50	0.41	700
3/8	6.00	0606	11.09	0.31	0.56	0.50	1,040
1/2	6.00	0806	12.69	0.38	0.63	0.75	1,800
5/8	6.00	1006	14.19	0.50	0.75	0.87	2,700
3/4	6.00	1206	15.81	0.63	1.00	1.00	4,000
1	6.00	1606	19.25	0.88	1.25	1.38	7,600
1	12.00	1612	25.25	0.88	1.25	1.38	7,600
1-1/4	12.00	2012	27.79	1.13	1.88	1.71	12,000

(1) PIN size is trade size (A) in 1/16 inch including leading zero if required and clear opening (B) in full inches including leading zero if required
(2) Dimension “C” is without jam nuts installed

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3.4.11.4 Type XVIII, class 3, eye and eye turnbuckle. The eye and eye turnbuckles shall be similar to that shown on figure 25, and shall be accordance with the dimensions and working loads shown in table XXVI.

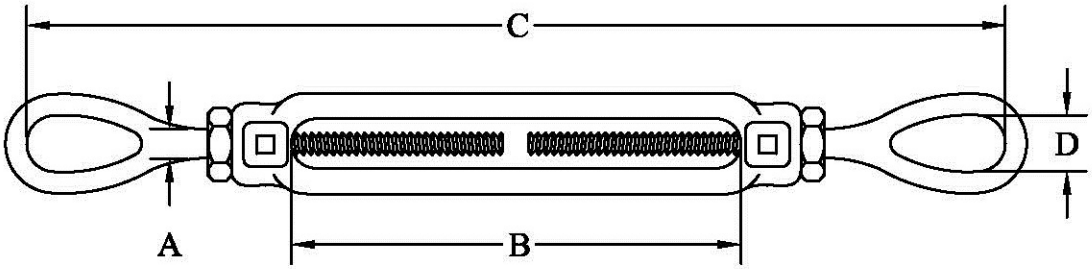


FIGURE 25. Type XVIII, class 3, eye and eye turnbuckle.

TABLE XXVI. Type XVIII, class 3, eye and eye turnbuckle.

Trade Size A (Thread Size) (nom) (inch)	Clear Opening B (min)(inch)	PIN size number(1)	Length C (max at full retraction)(2) (inch)	Eye width D (nom)(inch)	Working load (lbf)
1/4	4.00	0404	8.33	0.33	500
5/16	4.50	0504	9.66	0.41	800
3/8	6.00	0606	12.00	0.50	1,200
1/2	6.00	0806	13.75	0.75	2,200
5/8	6.00	1006	15.37	0.87	3,500
3/4	6.00	1206	17.25	1.00	5,200
1	6.00	1606	21.38	1.38	8,000
1	12.00	1612	26.88	1.38	8,000
1-1/4	12.00	2012	30.09	1.71	12,500

(1) PIN size is trade size (A) in 1/16 inch including leading zero if required and clear opening (B) in full inches including leading zero if required
(2) Dimension “C” is without jam nuts installed

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3.4.11.5 Type XVIII, class 4, stub and stub. The stub and stub turnbuckles shall be similar to that shown on figure 26, and shall be accordance with the dimensions and working loads shown in table XXVII.

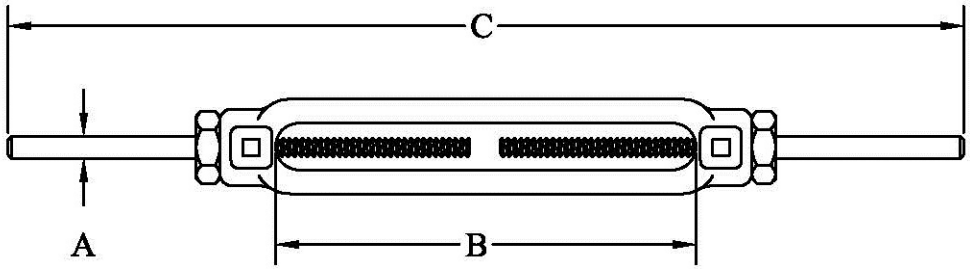


FIGURE 26. Type XIV, class 4, stub and stub.

TABLE XXVII. Type XIV, class 4, stub and stub.

Trade Size A (Thread Size) (nom) (inch)	Clear Opening B (min)(inch)	PIN size number(1)	Length C (max at full retraction)(2) (inch)	Working load (lbf)
1/4	4.00	0404	9.00	500
5/16	4.50	0504	10.50	700
3/8	6.00	0606	13.00	1,040
1/2	6.00	0806	14.13	1,800
5/8	6.00	1006	15.00	2,700
3/4	6.00	1206	16.00	4,000

(1) PIN size is trade size (A) in 1/16 inch including leading zero if required and clear opening (B) in full inches including leading zero if required
(2) Dimension “C” is without jam nuts installed

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3.4.11.6 Type XVIII, class 5, turnbuckle body. The turnbuckle body shall similar to that shown on figures 27, and shall be in accordance with the dimensions and working loads shown in table XXVIII. The turnbuckle body shall be supplied drilled and tapped for UNC/UNCR thread of trade size "A" per 3.4.13.1.2 unless otherwise specified in the order or contract.

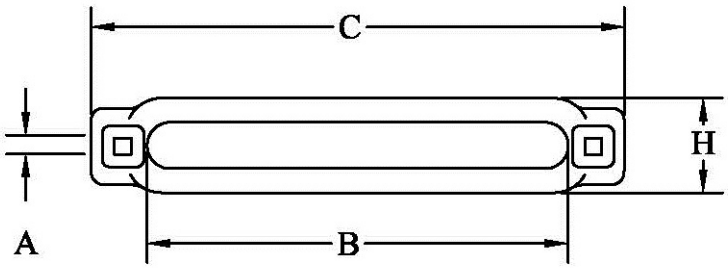


FIGURE 27. Type XVIII, class 5, turnbuckle body.

TABLE XXVIII. Type XIV, class 5, turnbuckle body.

Trade Size (Thread Size) A (nom) (inch)	Clear Opening B (min)(inch)	PIN size number(1)	Length C (min)(inch)	Body width H (nom)(inch)	Working load (lbf)
1/4	4.00	0404	4.75	0.72	500
5/16	4.50	0504	5.50	0.82	700
3/8	6.00	0606	7.13	0.97	1,040
1/2	6.00	0806	7.50	1.16	1,800
5/8	6.00	1006	7.81	1.38	2,700
3/4	6.00	1206	8.19	1.69	4,000
1	6.00	1606	9.00	2.31	7,600
1	12.00	1612	14.81	2.31	7,600
1-1/4	12.00	2012	15.13	2.62	12,000
1-1/2	12.00	2412	15.81	3.20	13,400

(1) PIN size is trade size (A) in 1/16 inch including leading zero if required and clear opening (B) in full inches including leading zero if required

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3.5 Workmanship. The chain and attachments shall be free from injurious imperfections that may impair appearance or serviceability. The forged and welded chain and attachments shall be free from fins, cracks, flaws, or seams. The screw threads shall be true to form, clean cut, and free from injurious defects.

3.6 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are for conformance inspection.

4.2 Lot.

4.2.1 Type I chains. For the purpose of inspection, a lot shall consist of not more than 3,000 feet of the same class and size chain, or any single continuous length of chain of over 3,000 feet. The chains shall have been manufactured using the same lot of materials and same production run, and offered for delivery at one time.

4.2.2 Attachments. For the purpose of inspection, a lot shall consist of not more than 10,000 attachments made from the same material, of the same type and class, offered for delivery at one time. Same material is defined as a single alloy, whether it is from a single heat or multiple heats.

4.3 Sampling procedures.

4.3.1 Chains.

4.3.1.1 Sampling for visual and dimensional inspection, type I chain. A sample of not less than 2 feet in length shall be selected from each lot of chain for visual and dimensional inspection. If lots comprise more than one length of chain, the sample links shall be selected equally from each chain.

4.3.1.2 Sampling for proof testing.

4.3.1.2.1 Type I chain. Type I, class 1, 2 and 3 chain shall be 100 percent proof tested. Type 1, class 5 chain does not require proof testing.

4.3.1.3 Sampling for break and elongation testing.

4.3.1.3.1 Type I chain. For chain 7/32” and smaller a minimum of 9 links per lot are to be used in test specimen. For chain larger than 7/32” but less than 3/4” a minimum of 7 links per lot are to

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be used in the test specimen. For chain 3/4" and larger a minimum of 3 links per lot are to be used in the test specimen.

4.3.2 Attachments.

4.3.2.1 Accept at zero sampling plans. Except as noted below or in the contract (see 6.2), sampling shall be to an "accept at zero" (c=0) sampling plan such as MIL-STD-1916 at verification level IV or ASQ H1331 at Index value (associated AQL) of 1.0.

4.3.2.2 Sampling for inspection and proof load. For all attachments, except type II connecting links and type III lap links, a random sample of attachments for visual inspection and proof load shall be selected from each lot in accordance with 4.3.2.1.

4.3.2.3 Sampling for breaking and ductility tests. For all attachments except type II connecting links and type III lap links, a random sample of attachments shall be selected in accordance with table XXIX for the breaking and ductility tests specified in 4.4.2.

TABLE XXIX. Sampling for destructive testing of attachments.

Number of attachments per lot	Number to be selected for breaking load and ductility tests
1 to 8	1
9 to 50	2
51 to 500	3
501 to 10,000	5

4.3.2.4 Type II and type III links. For visual inspection of type II connecting links and type III lap links, a random sample of attachments shall be selected from each lot in accordance with 4.3.2.1 except a minimum of 2 type II links (4 link halves) are required for interchangeability testing. For proof and break testing, a sample of attachments shall be randomly selected in accordance with table XXIX. For type II links, 2 link halves are required for each test link. The proof and break tests may be performed on the same sample lot.

4.4 Test methods.

4.4.1 Chains.

4.4.1.1 Order of tests. Testing shall be performed in the following order:

a. Chemical analysis
b. Visual and dimensional inspection
c. Proof test
d. Breaking test
e. Elongation test (type I chain only).

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4.4.1.1.1 Chemical analysis. Chemical analysis of the stainless steel shall be determined from the certified mill data sheet, by analysis of the raw stainless steel sampled from the stainless steel heat, or by analysis of samples of finished chain.

4.4.1.1.2 Examination of product. Each sample of chain selected as specified in 4.3.2.1 shall be examined to determine conformance to this specification regarding dimensions, finish, quality of welds, and other requirements. The chains shall match the dimensions specified in the respective tables.

4.4.1.1.3 Fixturing for proof and break test. The sample shall be free from twist. The holding arrangement shall have the stresses bearing on the end links of the test sample the same as on the balance of the links.

4.4.1.1.4 Measurements for elongation. The chain sample shall be measured to the inside of the end links after the chain has been proof tested. In determining the length of chain, a load not exceeding 20 percent of the WLL shall be applied to take up the slack.

4.4.1.1.5 Proof tests. Chain shall be proof tested by subjecting it to the proof load specified in the respective section. After proof test, the chain shall be thoroughly inspected. If any links show broken welds, surface rupture, inferior workmanship or other defects, the entire lot shall be rejected. Point contact marks shall not be considered as deformations.

4.4.1.1.5.1 Type I, classes 1, 2 and 3 chain. 100% of all type I, class 1, 2 and 3 chain will be subjected to the proof tests.

4.4.1.1.5.2 Type I, class 5 chain. Type I, class 5 chain does not require separate proof testing as fabrication of twist verifies strength of link.

4.4.1.1.6 Breaking test. Select specimens per 4.3.1.3 from each lot of chain shall be subjected to the breaking test. The chain shall be considered to have passed the breaking test if it withstands the specified load or if it breaks at a load greater than that specified in the respective section.

4.4.1.1.7 Elongation test type I chain. The elongation measurement shall be made on the remains of the breaking test sample. The actual gauge length shall be determined before the proof test. After the breaking test the gauge shall be measured to determine the amount of elongation. The elongation shall be not less than 15 percent.

4.4.1.2 Rejections. If any sample fails the chemical, proof, breaking, or elongation tests, the lot shall be rejected. Failure of any other test or examination requires 100% inspection for acceptance of the lot.

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4.4.2 Attachments.

4.4.2.1 Order of tests. Testing shall be performed in the following order:

a. Chemical analysis
b. Visual and dimensional inspection
c. Proof test
d. Breaking test
e. Ductility test

4.4.2.1.1 Chemical analysis. Chemical analysis of the stainless steel shall be determined from the certified mill data sheet, by analysis of the raw stainless steel sampled from the stainless steel heat, or by analysis of samples of finished attachment.

4.4.2.1.2 Examination of product. A random sample of attachments as specified in 4.3.2 shall be examined to determine conformance to this specification regarding dimensions, finish, quality of welds, and other requirements.

4.4.2.1.2.1 Type II connecting link interchangeability. The links selected for proof and break load testing shall be visually inspected to ensure that all links are interchangeable without applying force. The two halves shall fit concentrically to each other and shall mate (face flush fit) to each other maintaining rivet-pin to hole alignment.

4.4.2.1.2.2 Type II connecting link rivet size. The rivets shall be of sufficient diameter and length to completely fill the corresponding holes when peened. Rivet size shall be checked on samples assembled for proof and break testing.

4.4.2.1.3 Fixturing for proof and break tests. For all attachments with round loading points, including but not limited to any type V hooks with an eye, type VI rings and type XVI master links, the fixture shall be a maximum of 60% of the bearing diameter and large enough to prevent localized point loading. Bearing diameter is the ring ID or equivalent dimension and not the material diameter. When applicable, samples may be tested in series.

4.4.2.1.4 Proof test. Samples selected as specified in 4.3.2 shall be subjected to proof loads. Before testing, all dimensions shall be measured and recorded. After applying the test load, the attachments shall be measured and the samples shall be considered satisfactory if deformation does not exceed 2% except as noted. The proof load shall be as specified in the respective section. Point contact marks shall not be considered as deformations.

4.4.2.1.4.1 Type V hooks. Hooks shall withstand the proof test with no more than 1% permanent opening of the hook at the throat. Deformation of all other dimensions shall not exceed 2%.

4.4.2.1.4.2 Type VI rings, type XIII pear-shaped links and type XVI master links. Type VI rings, type XIII pear-shaped links, and type XVI master links are exempt from the deformation requirement.

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4.4.2.1.5 Breaking test. After proof testing, a subset of samples selected as specified in 4.3.2 shall be subjected to the minimum breaking load as specified in the respective section. The attachment will be considered to have passed the breaking test if it withstands the specified load or if it breaks at a load greater than specified.

4.4.2.1.6 Ductility test. The ductility measurement shall be made on the remains of the breaking test sample. The actual dimensions of the attachment shall be determined before the breaking test. After the breaking test the attachment shall be measured to determine the amount of elongation. The ductility shall be not less than 15 percent or as otherwise specified.

4.4.2.2 Rejections. If any sample fails a test or examination that prevents use of the sample, the lot shall be rejected. This includes but is not limited to any break test or proof test of connecting links or lap links. If any sample fails any other test or examination, the lot may be 100% inspected and resubmitted.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. This specification defines the requirements for a variety of corrosion resistant steel (CRES) chains and attachments. Many of these are CRES variants of carbon and alloy steel items defined in RR-C-271.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. PIN to include type and class of chain or attachments required (see1.2 and 1.3).
c. Material if required (see 3.1 and 3.1.3)
d. Finish other than that specified (see 3.2.1 and 3.2.2).
e. Size of chain or attachments required (see applicable tables).
f. Screw-pin and bolt shackle thread type (see 3.5.2).
g. Packaging requirements (see 5.1).

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6.3 Subject term (key word) listing.

Breaking load
Connecting link
Hook
Lap link
Lifting chain
Long-link chain
Master link
Mechanical coupling link
Pear link
Proof load
Rectangular link
Round ring
Sash chain
Shackle
Triangle loop
Turnbuckle
Twist link

Custodians:

Navy - SH
Air Force - 99
DLA - GS

Review Activities:
Air Force - 71, 11
Other - DS

Preparing Activity:
DLA - GS5

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